

We use the setting and the notation of Tutorial 1: $K \geq 2$ arms with distributions $\nu_1, \dots, \nu_K$ on $[0, 1]$ and means $\mu_1, \dots, \mu_K$, best mean $\mu^* = \max_a \mu_a$, gaps $\Delta_a = \mu^* - \mu_a$, horizon $T > K$, numbers of pulls $N_a(t)$ and empirical means $\hat{\mu}_a(t)$. The strategy UCB1 plays $A_t = t$ for $t \in \{1, \dots, K\}$ and, for $t \in \{K+1, \dots, T\}$,

$$A_t = \min \arg \max_{1 \leq a \leq K} U_a(t) \quad , \quad \text{where} \quad U_a(t) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{N_a(t-1)}} \quad , \quad (1)$$

so that ties go to the smallest index.

Exercise 1 (UCB1 without noise). Assume $K = 2$ and that the rewards are deterministic: ν_a is the Dirac mass at μ_a for $a \in \{1, 2\}$, with $\mu_1 > \mu_2$. We write $\Delta = \mu_1 - \mu_2$.

1. Show that $\hat{\mu}_a(t) = \mu_a$ for any $a \in \{1, 2\}$ and $t \geq 2$. Deduce the value of $N_2(T)$ for the greedy strategy (Tutorial 1, equation (5)), and its regret.
2. Let $t \in \{3, \dots, T\}$ be a round at which UCB1 plays $A_t = 2$. Show that

$$\sqrt{\frac{2 \log t}{N_2(t-1)}} > \Delta \quad , \quad \text{and deduce that} \quad N_2(t-1) < \frac{2 \log t}{\Delta^2} \quad .$$

3. Deduce that, for UCB1,

$$N_2(T) < \frac{2 \log T}{\Delta^2} + 1 \quad . \quad (2)$$

Hint: consider the last round at which arm 2 is played.

4. Show that, for UCB1, $N_2(T) \rightarrow \infty$ as $T \rightarrow \infty$.
Hint: assume that arm 2 is pulled n times in total, and compare the limits of $U_1(t)$ and $U_2(t)$ as $t \rightarrow \infty$.
5. Assume $\Delta = 0.2$ and $T = 5000$. Compute the right-hand side of (2) and the corresponding bound on the regret of UCB1, and compare it with the regret of greedy.
Hint: $\log 5000 \approx 8.52$.

Exercise 2 (Number of pulls of a suboptimal arm). Let a^* be an arm such that $\mu_{a^*} = \mu^*$ and let a be an arm such that $\Delta_a > 0$. For $t \in \{K+1, \dots, T\}$, consider the two inequalities

$$\mu^* \leq U_{a^*}(t) \quad \text{and} \quad \hat{\mu}_a(t-1) \leq \mu_a + \sqrt{\frac{2 \log t}{N_a(t-1)}} \quad . \quad (3)$$

1. Let $n \geq 1$ and $t \geq 2$ be integers, and let $Y_1, \dots, Y_n$ be i.i.d. with values in $[0, 1]$ and mean μ . Show that

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n Y_i \geq \mu + \sqrt{\frac{2 \log t}{n}}\right) \leq \frac{1}{t^4} \quad .$$

Hint: use Hoeffding's inequality (Tutorial 1, equation (6)).

2. Let $t \in \{K+1, \dots, T\}$. Show that if $A_t = a$, then $U_a(t) \geq U_{a^*}(t)$.
3. Let $t \in \{K+1, \dots, T\}$ be such that $A_t = a$ and such that the two inequalities of (3) hold. Show that

$$\Delta_a \leq 2 \sqrt{\frac{2 \log t}{N_a(t-1)}} \quad , \quad \text{and deduce that} \quad N_a(t-1) \leq \frac{8 \log t}{\Delta_a^2} \quad .$$

4. Assume that the two inequalities of (3) hold for any $t \in \{K + 1, \dots, T\}$. Show that

$$N_a(T) \leq \frac{8 \log T}{\Delta_a^2} + 1 . \quad (4)$$

Hint: consider the last round at which arm a is played.

5. Assume $K = 2$, $\Delta_2 = 0.2$ and $T = 5000$. Compute the right-hand side of (4) and the corresponding bound on $\Delta_2 N_2(T)$. Compute the factor by which $8 \log(T)/\Delta_a^2$ is multiplied when Δ_a is replaced with $\Delta_a/2$, then when T is replaced with T^2 .

Hint: $\log 5000 \approx 8.52$.

Exercise 3 (UCB1 by hand on three arms). Assume $K = 3$ and that the first six rewards observed by UCB1 are

$$X_1 = 1 , \quad X_2 = 0 , \quad X_3 = 1 , \quad X_4 = 0 , \quad X_5 = 1 , \quad X_6 = 0 . \quad (5)$$

The following approximate values may be used, together with $1/\sqrt{2} \approx 0.707$ and $1/\sqrt{3} \approx 0.577$.

t	4	5	6	7
$\sqrt{2 \log t}$	1.665	1.794	1.893	1.973

1. Compute $U_a(4)$ for $a \in \{1, 2, 3\}$, and deduce A_4 .
2. Compute $N_a(4)$, $\hat{\mu}_a(4)$ and $U_a(5)$ for $a \in \{1, 2, 3\}$, and deduce A_5 .
3. Compute A_6 and A_7 .
4. Let $G_7 = \min \arg \max_a \hat{\mu}_a(6)$ be the arm that the greedy strategy would play at round 7 after the same history. Compute G_7 , and compute the smallest value of $N_2(6)$ for which one would have $U_2(7) < U_1(7)$, the other quantities being unchanged.